



Transistor-Level Design and Physical Implementation of a 10-Bit Linear Feedback Shift Register (LFSR)

Digital IC Design and Tape out - ED221
B.Tech Electronics and VLSI Design and Technology
Instructor - Prof. Biswajit Mishra Sir

Group Members

Vishv Gorasiya - 202404045
Rudrapratapsinh Rathod - 202404053
Jeet Sakariya - 202404036
Kuldip Solanki - 202404040

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I am also thankful to Dhirubhai Ambani University (Formerly DA-IICT) for providing the necessary laboratory infrastructure, Cadence EDA tool access, and the SCL 180nm PDK resources required to successfully execute and verify this silicon design.

Abstract

This report documents the full-custom, transistor-level physical design and post-silicon characterization of a 10-bit Linear Feedback Shift Register (LFSR) in the SCL 180nm CMOS process node. To maintain stringent control over device geometry, active-area sharing, and parasitic routing penalties, automated standard-cell synthesis methodologies were intentionally bypassed. The project encompasses the complete Backend-of-Line (BEOL) workflow: schematic capture of fundamental logic primitives (NAND, XOR), custom standard cell layout utilizing Euler path optimization, top-level datapath integration, and Siemens Calibre sign-off (DRC, LVS, and PEX). Post-layout RC parasitic extraction confirms a highly optimized physical silicon layout, achieving a maximum operating frequency of 1.75 GHz and a dynamic power footprint of 246.1 μW at a nominal 1.8V supply, validating the macro for high-performance, mixed-signal SoC integration.

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1. Introduction

1.1 Objective and Scope

The primary objective of this project is the full-custom, transistor-level physical implementation of a 10-bit Linear Feedback Shift Register (LFSR). By intentionally bypassing automated logic synthesis, this design enforces strict manual control over device sizing and backend-of-line (BEOL) routing. The goal is to extract the physical RC parasitics of the completed layout to accurately quantify the delay penalties and determine the maximum operating frequency (F_{max}) and dynamic power of the silicon macro.

1.2 The Universal NAND Approach

The entire LFSR architecture was synthesized using a universal static-CMOS NAND topology. NAND gates are utilized over NOR gates due to the higher mobility of electrons, which allows for better-balanced rise and fall transition times. By exclusively using NAND-based logic to construct the custom XOR feedback network and the 32-transistor D-Flip-Flops, the design eliminates the threshold voltage (V_{th}) degradation and localized clock-skew vulnerabilities common in pass-transistor logic, guaranteeing full 1.8V rail-to-rail signal integrity.

1.3 EDA Tools and SCL 180nm Technology Node

- **Technology Node:** SCL 180nm node (Nominal $V_{DD} = 1.8V$).
- **Schematic & Layout:** Cadence Virtuoso Layout Suite.
- **Sign-Off Verification:** Calibre for Design Rule Checking (DRC) and Layout Versus Schematic (LVS) matching.
- **Extraction & Simulation:** Calibre xRC for parasitic extraction (PEX) and Cadence Spectre (ADE Explorer) for post-layout AC/DC characterization.

2. Theoretical Foundation: LFSR Topology and Tap Selection

Before detailing the transistor-level implementation, it is critical to define the mathematical and structural primitives that govern the 10-bit LFSR design.

2.1 Mechanical Definition of the LFSR

A Linear Feedback Shift Register (LFSR) is a synchronous, discrete-time state machine. Mechanically, it is a standard shift register where the deterministic input to the first stage is a linear function of its previous overall state. By using an exclusive-OR (XOR) gate to calculate parity across specific tap points in the register, the system generates a Pseudo-Random Binary Sequence (PRBS) that exhibits high statistical randomness while remaining completely deterministic.

2.2 Hardware Primitives for a 10-Bit LFSR

To construct this architecture in physical silicon, three fundamental primitives are required:

1. **Sequential Storage:** A cascade of **ten positive-edge-triggered D-Flip-Flops** to hold the current 10-bit state vector.
2. **Combinational Feedback:** A modulo-2 adder (**XOR gate**) to calculate the feedback logic.
3. **Non-Zero Initialization Seed:** A hardwired state injection (utilizing Preset/Reset limits) to prevent the LFSR from entering a permanent "zero-lock" state, as an all-zero vector fed into an XOR network will perpetually output zero.

2.3 Tap Selection: The x10 and x7 Polynomial

The specific sequence generated by the LFSR is dictated by its characteristic polynomial over a **Galois Field, GF(2)**. For a 10-bit system, the goal is to achieve a Maximal Length Sequence (MLS), meaning the register will cycle through every

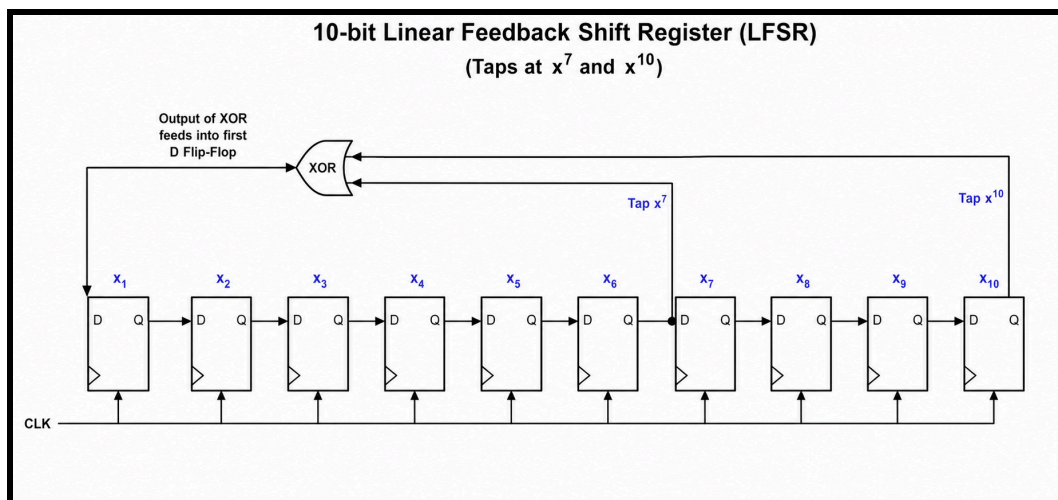
possible non-zero state exactly once before repeating. **The period of an MLS is $2N-1$, which yields 1023 unique states for this 10-bit design.**

To achieve an MLS, the feedback polynomial must be a **primitive polynomial**. For this design, I explicitly selected the following polynomial :

$$P(x) = x^{10} + x^7 + 1$$

This selection was not arbitrary. The mathematical taps directly translate to physical hardware connections:

- x^{10} : dictates a tap from the final output stage (Q10).
- x^7 : dictates a tap from the seventh output stage (Q7).
- **The Hardware Advantage:** While other primitive polynomials exist for a 10-bit system (such as $x^{10}+x^3+1$), many require more than two tap points (e.g., $x^{10}+x^8+x^3+x^2+1$). Every additional tap requires cascaded XOR gates in the feedback loop. By choosing a trinomial (a polynomial with exactly three terms), the feedback network requires only a **single 2-input XOR gate**. This architectural choice drastically reduces the combinational propagation delay (tXOR) on the critical path, directly maximizing the absolute maximum operating frequency (Fmax) of the physical silicon.



3. Sequential Logic Architecture: The D-Flip-Flop

3.1 Topological Comparison: Master-Slave vs. 74 series Logic

The sequential backbone requires a robust positive-edge-triggered D-Flip-Flop. When synthesizing at the static-CMOS gate level, the **74-series 6-NAND topology** is vastly superior to a conventional 8-NAND Master-Slave architecture for high-frequency operations.

- **The 8-NAND Master-Slave Vulnerability:** Cascading two level-sensitive latches requires internal complementary clock phases (CLK and CLK'). Generating this local inversion introduces intra-cell clock skew, degrading the metastability window (setup time, t_{su} , and hold time, t_h) and increasing the probability of internal race conditions at gigahertz frequencies.
- **The 6-NAND Edge-Triggered Solution:** This LFSR utilizes the **7474-style architecture**, constructed from **three cross-coupled SR latches**. Specifically, this topology is built using exactly **four 3-input NAND gates** and **two 2-input NAND gates** (totaling **32 transistors**). This configuration operates strictly on a single-ended global clock. By completely eliminating localized clock inverters, it eradicates internal clock skew, minimizes Clock-to-Q propagation delay (t_{cq}), and directly maximizes f_{max} .

3.2 Asynchronous State Initialization: Preset and Reset Variants

XOR-based LFSRs are critically susceptible to a permanent **"Zero-Lock"** state; an all-zero initial vector will perpetually output logic '0'. **Injecting a non-zero seed via datapath multiplexers introduces combinational parasitic delay, which severely degrades the maximum operating frequency.**

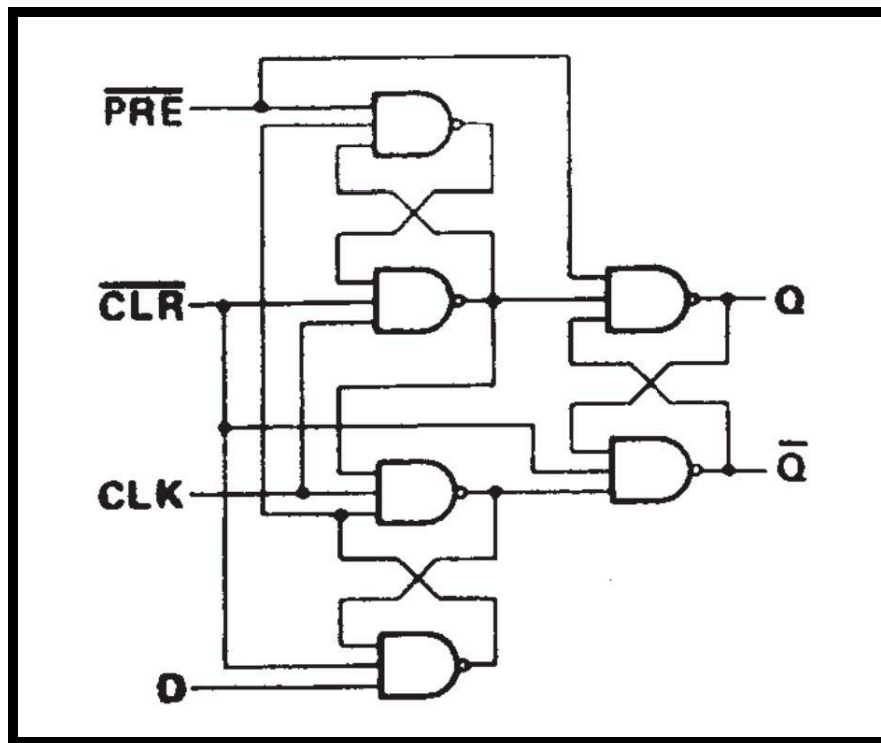
To bypass this critical path penalty, hardware-level initialization is executed utilizing two custom variants of the **6-NAND standard cell**, integrating asynchronous active-low controls directly into the specific **2-input and 3-input NAND gates** of the internal SR latches:

- **Dff7474_prst (Preset Variant):** Forces the internal slave latch to VDD (logic '1') upon global system reset.
- **Dff7474_rst (Reset Variant):** Forces the internal master latch to GND (logic '0') upon global system reset.

By physically alternating the instantiation of these two cells across the 10-bit shift register chain, a deterministic, non-zero pseudo-random seed is instantaneously hardwired into the silicon during power-up, requiring zero combinational overhead on the synchronous datapath.

3.3 Reference Architecture and Custom Deviation

The theoretical baseline for this design is the **standard 7474 D-Flip-Flop**, which utilizes a dual-control 6-NAND topology featuring both **asynchronous Preset (PRE)** and **Clear (CLR)** networks.

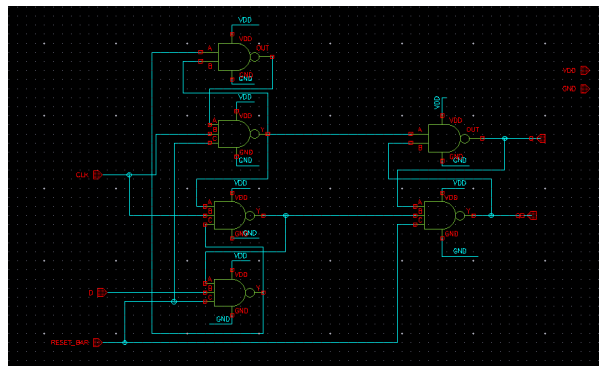


*[6] Texas Instruments, "SN7474 Dual D-Type Positive-Edge-Triggered Flip-Flops with Preset and Clear," Product Datasheet. (Reference for the 6-NAND dual-control sequential baseline architecture).

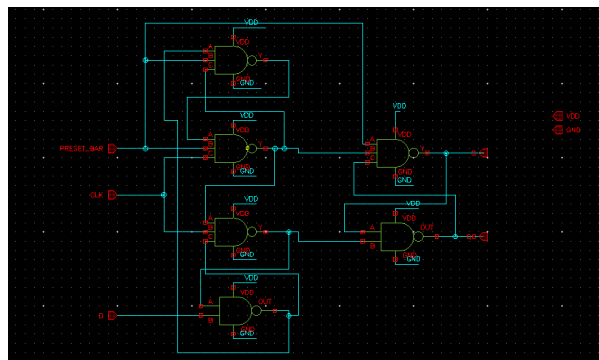
Deploying this universal cell across the entire LFSR is highly inefficient. To hardwire an initialization seed, a specific register stage only requires a logic '1' OR a logic '0' upon power-up—never both. To minimize silicon real estate and parasitic loading, the reference architecture was bifurcated into two application-specific standard cells:

- **Dff7474_prst:** The CLR network was entirely removed. The associated 3-input NAND gates were downgraded to 2-input NANDs, significantly reducing gate capacitance (C_g) and physical cell area.
- **Dff7474_rst:** The PRE network was similarly stripped, retaining only the asynchronous logic intersecting the master latch.

This architectural deviation aggressively cuts unnecessary devices. By stripping the redundant control logic, the custom cells alleviate BEOL routing congestion and minimize parasitic drive requirements while preserving deterministic state injection.



Dff7474_rst(Reset)

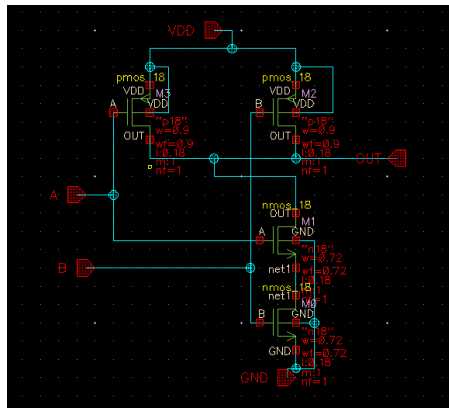


Dff7474_prst(Preset)

4. Circuit Design and Transistor-Level Implementation

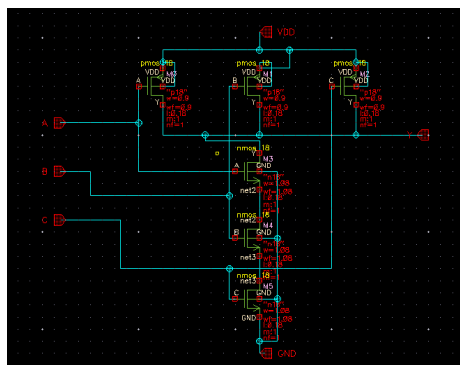
4.1 Combinational Primitives

- **2-Input NAND (NAND2):** 4-transistor topology (2 parallel PMOS, 2 series NMOS). To equalize mobility differences between holes and electrons ($\mu_n \approx 2.5\mu_p$) and achieve symmetrical rise/fall times ($t_r \approx t_f$), the PMOS width (W_p) was scaled to approximately 2.5 times the NMOS width (W_n).



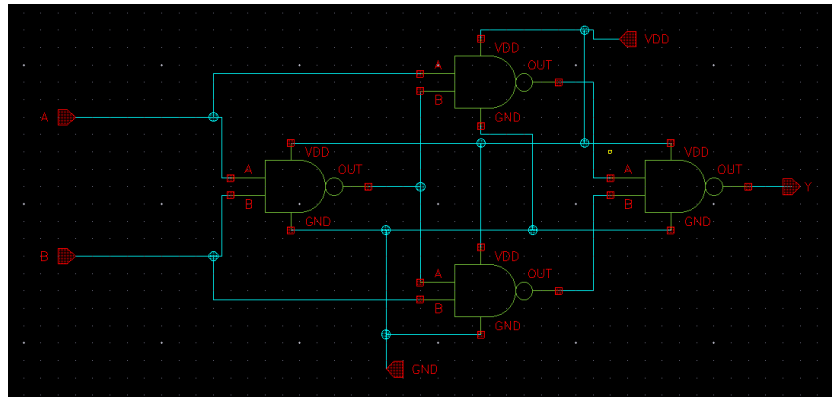
NAND2_Schematic

- **3-Input NAND (NAND3):** 6-transistor topology. The series NMOS stack was upsized to mitigate body-effect degradation and increased on-resistance (R_{on}), ensuring the pull-down network meets high-frequency timing constraints.



NAND3_Schematic

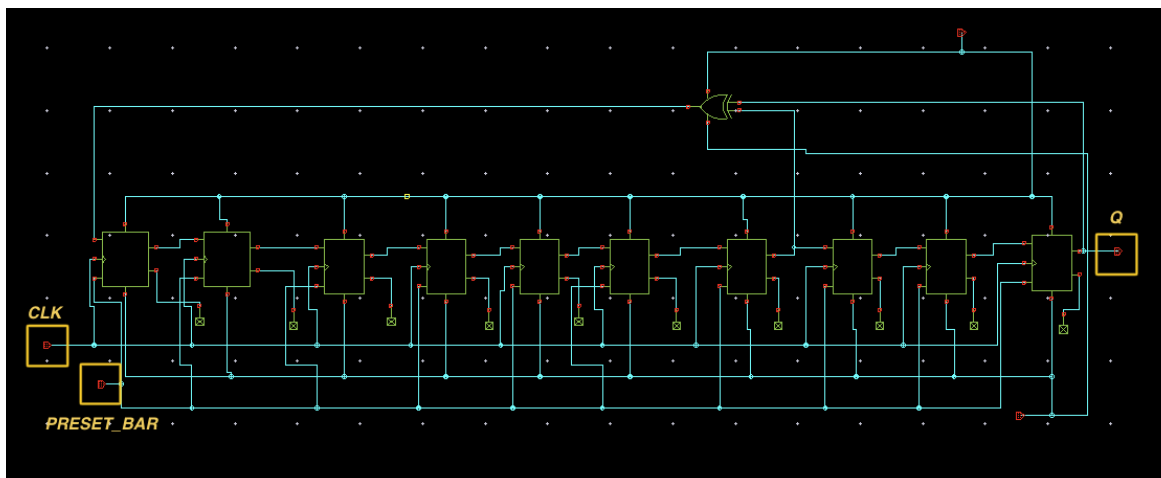
- **Modulo-2 Adder (XOR):** 16-transistor topology constructed from four interconnected NAND2 gates. This bypasses pass-transistor logic (PTL), preventing threshold voltage (V_{th}) drops and guaranteeing a full 1.8V output swing without requiring secondary restoring buffers.



XOR_Schematic

4.2 Top-Level LFSR Integration and Datapath

The 10-bit macro is integrated via three primary routing networks:



10-bit_LFSR_Schematic

1. **Sequential Pipeline:** The ten D-Flip-Flops are cascaded serially ($D_{n+1}=Q_n$).
2. **Combinational Feedback:** Tap points Q7 and Q10 are routed to the 16T XOR gate. The XOR output drives D1.
3. **Hardwired Seed Injection:** A global active-low preset net (**PRESET_BAR**) drives all asynchronous control pins. To achieve the specific **1100000000** initialization vector, the physical layout dictates that the first two stages (Q1,Q2) are populated with **Dff7474_prst** cells, while the remaining eight stages (Q3 through Q10) are populated with **Dff7474_rst** cells. Asserting **PERST_BAR** instantaneously hardwires this deterministic seed directly into the physical latches.

LFSR Device Integration Breakdown :

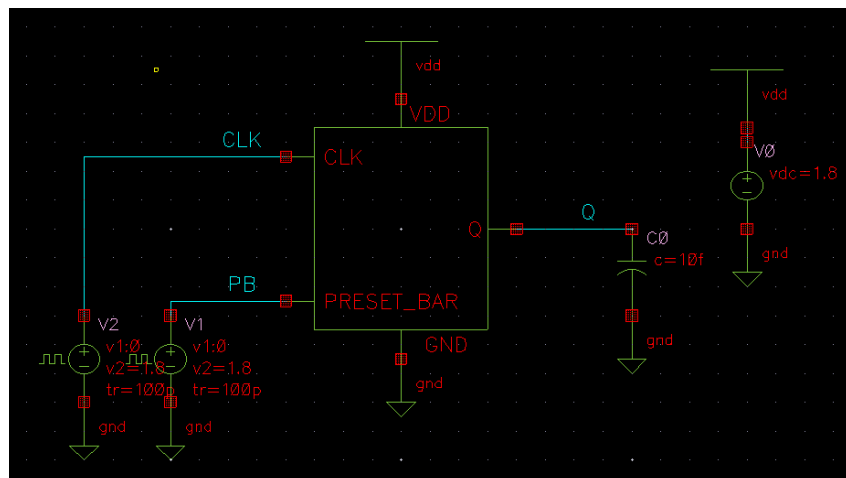
Standard Cell	Composition	Instance Count	Total Transistors
NAND2	4 Transistors / Gate	24 Total Gates	96
Total NAND3 Primitives	6 Transistors / Gate	40 Total Gates	240
Modulo-2 Adder (XOR)	4× NAND2	1 Logic Block	16
Dff7474_prst (Preset)	4× NAND3, 2× NAND2	2 (Stages 1&2)	64
Dff7474_rst (Reset)	4× NAND3, 2 ×NAND2	8 (Stages 3-10)	256
10-Bit LFSR	--	64 Total Primitives	336 Devices

5. Pre-Layout Simulation and Schematic Verification

Before committing to physical layout generation, the raw transistor-level netlists were simulated in Cadence ADE Explorer to verify logical correctness, initialization behavior, and baseline ideal timing characteristics.

5.1 Standard Cell Layout and Area Metrics

To confirm the mathematical accuracy of the combinational feedback loop, a transient analysis was executed over a 2 μs simulation window. The primary objective was to verify that the $P(x) = x^{10} + x^7 + 1$ XOR tap configuration successfully generated a maximal length **Pseudo-Random Binary Sequence (PRBS)**.



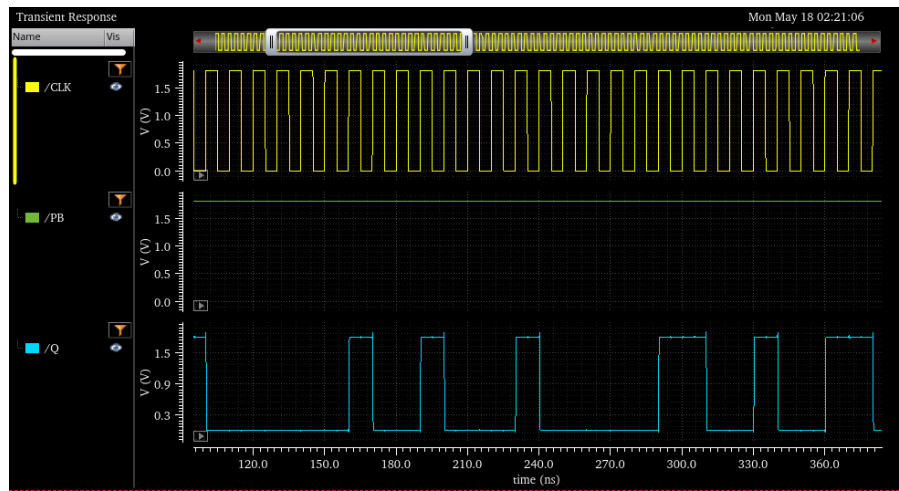
LFSR DUT

The simulation confirmed that the 10-bit register successfully cycles through exactly $2^{10} - 1 = 1023$ **unique states** before the sequence repeats. There are no dead-ends or sub-cycles in the datapath. The XOR gate strictly maintains a full 1.8V output swing without any threshold voltage degradation, confirming the robustness of the static-CMOS 4-NAND topology over pass-transistor logic.

5.2 Deterministic Seed Injection (Asynchronous Preset Verification)

The hardware-level initialization strategy was verified by forcing the global **PRESET_BAR** net to GND (0V) for 2 ns during the initial power-up phase.

Transient waveforms confirmed that asserting **PRESET_BAR** instantaneously bypassed the synchronous clock network. The two **Dff7474_prst** cells (Stages 1 and 2) were immediately forced to VDD (1.8V), while the eight **Dff7474_rst** cells (Stages 3 through 10) were simultaneously pulled to GND (0V) by this exact same global control signal. This successfully injected the strict **1100000000** initialization seed into the datapath, permanently preventing the LFSR **"Zero-Lock"** state without requiring slow combinational multiplexers on the critical path.



LFSR_Pre_Layout Simulation

5.3 Baseline AC Timing (Ideal Extraction)

Parameter	Description	Value (Ideal)
t_cq	Clock-to-Q Propagation Delay	213.1 ps
t_XOR	Combinational Delay	71.4 ps
t_min_ideal	Minimum Theoretical Clock Period	284.5 ps
f_max_ideal	Theoretical Maximum Operating Frequency	3.51 GHz

6. Physical Implementation and Layout Dimensions

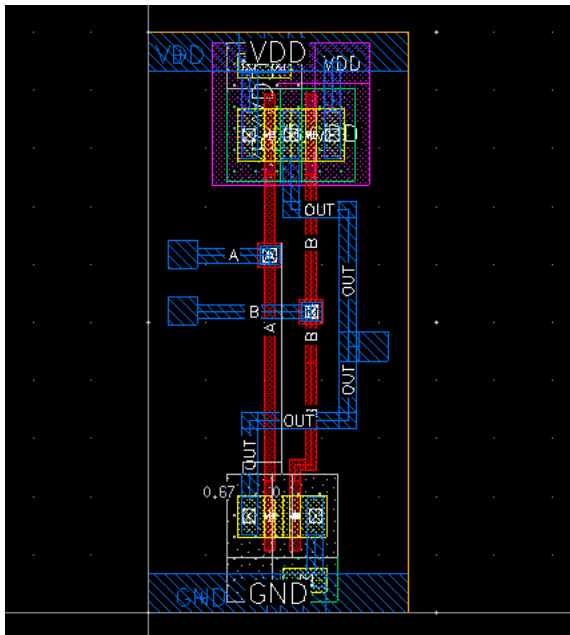
The custom layout was executed in Cadence Virtuoso using a hierarchical, bottom-up design methodology. All physical dimensions and routing networks were strictly optimized to minimize backend-of-line (BEOL) parasitics.

6.1 Standard Cell Layout and Area Metrics

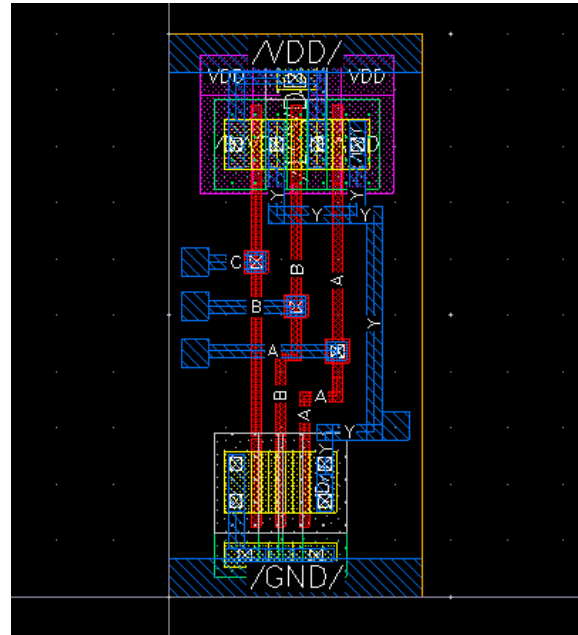
The physical bounding boxes for the fundamental logic gates, intermediate macros, and the final integrated system are broken down below.

6.1.1 NAND2 and NAND3 Primitives

- **Height (Y):** 10.0 μm
- **Width (X):** 4.5 μm
- **Calculated Area:** 45.0 μm^2



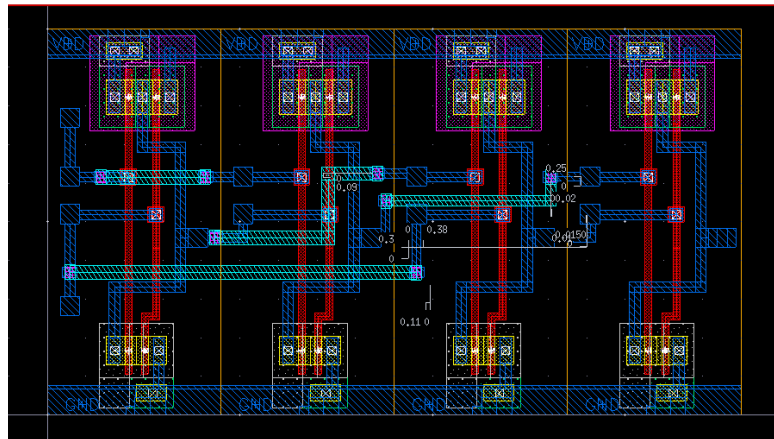
NAND2 Layout



NAND3 Layout

6.1.2 Modulo-2 Adder (XOR)

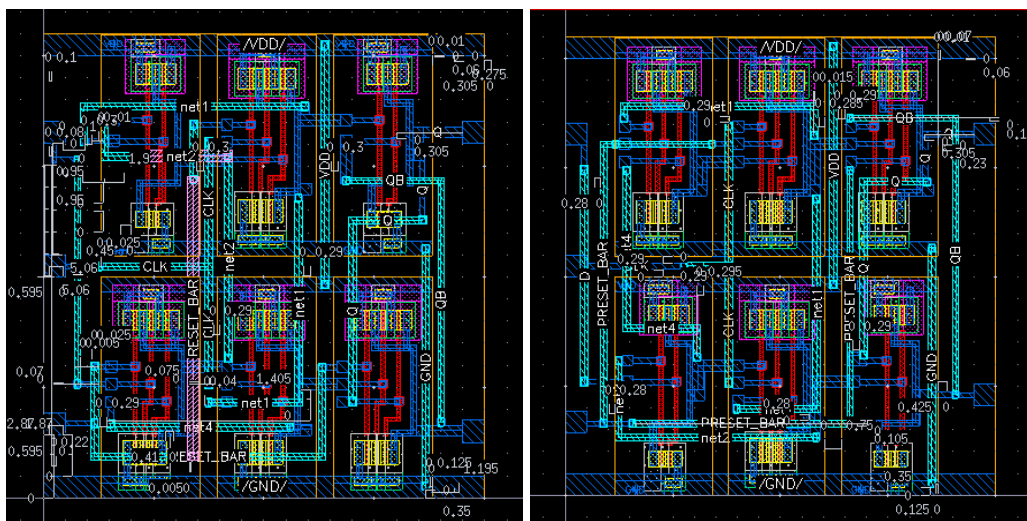
- **Height (Y):** 10.0 μm
- **Width (X):** 18.0 μm
- **Calculated Area:** 180.0 μm^2



XOR Layout

6.1.3 Sequential Elements

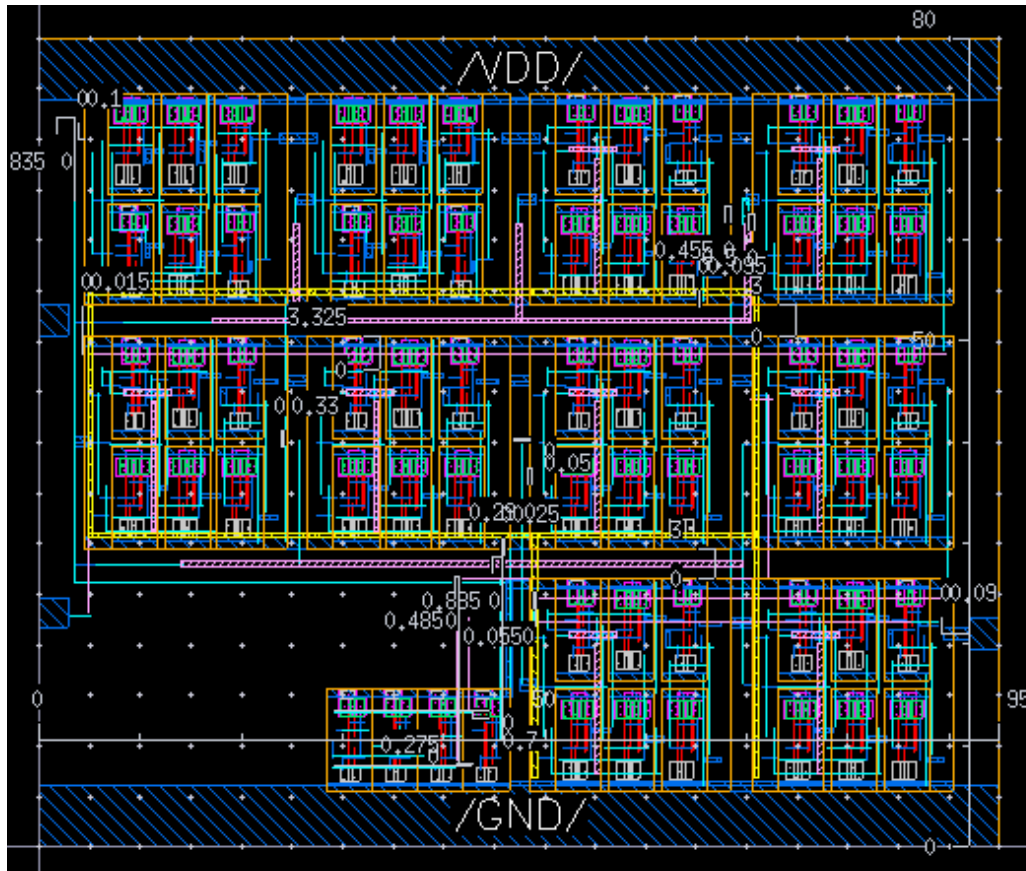
- **Height (Y):** 21.0 μm
- **Width (X):** 20.0 μm
- **Calculated Area:** 420.0 μm^2



Dff7474_rst(Reset) & Dff7474_prst(Preset) Layouts

6.1.4 Complete 10-Bit LFSR

- **Height (Y):** 80.0 μm
- **Width (X):** 95.0 μm
- **Total Silicon Area:** 7855.28 μm^2



10-bit LFSR Layouts

6.2 Physical Verification and Metal Density Constraints

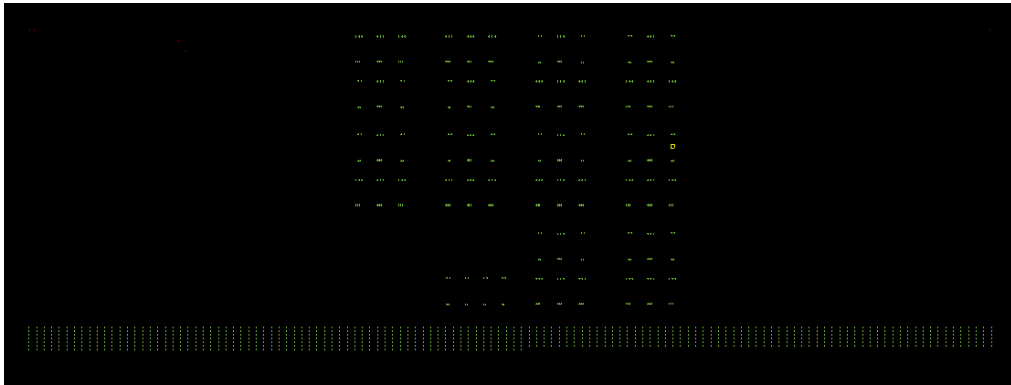
The extracted macro is 100% DRC and LVS clean in Siemens Calibre against the SCL 180nm ruleset (zero spacing, width, or enclosure violations).

Chemical Mechanical Polishing (CMP) dummy metal-fill density errors were deliberately waived. As this LFSR is an IP macro designed for higher-level integration, satisfying global density rules at this localized module stage artificially inflates the physical area and blocks necessary routing resources. The layout was optimized entirely for the minimal **7855.28 μm^2** footprint and clean signal routing.

7. Post-Layout Simulation and Characterization

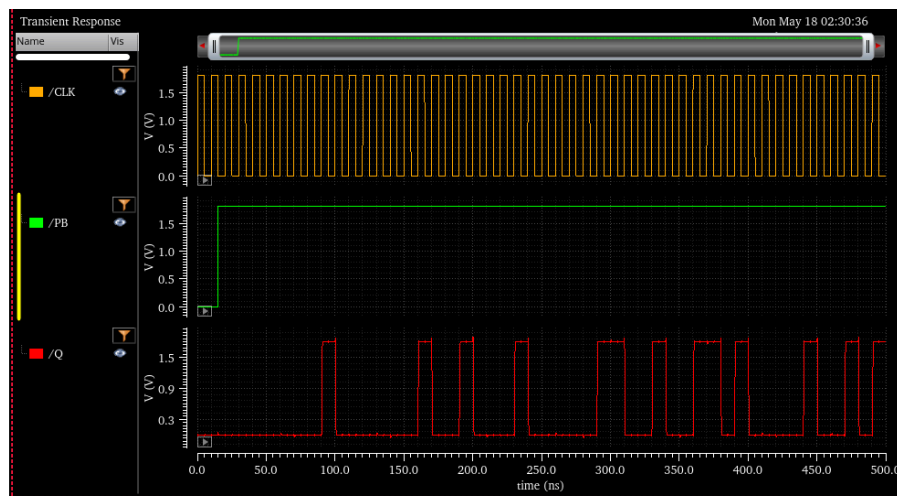
7.1 Parasitic Extraction (PEX) Methodology

Before **AC/DC** characterization, **Parasitic Extraction (PEX)** was performed using Siemens Calibre PEX. The physical layout was analyzed to extract the distributed Resistance and Capacitance (RC) network across all backend-of-line (BEOL) routing layers, device junctions, and substrate boundaries. This process generated a highly accurate, layout-annotated netlist.



Parasitic Extraction

7.2 Timing and Parasitic Degradation



Post Layout Simulation

The inclusion of physical routing parasitics introduced a mathematically expected delay penalty. Due to interconnect resistance and parasitic gate/diffusion capacitance (Cdb, Cgs), the **Clock-to-Q delay** increased by **73.2%** compared to ideal schematic simulations.

Parameter	Description	Pre-Layout (Ideal)	Post-Layout (PEX)
t_cq	Clock-to-Q Delay	213.1 ps	369.2 ps
t_su	Setup Time	N/A	78.5 ps
t_h	Hold Time	N/A	14.2 ps
t_rise	Output Rise Time	N/A	84.3 ps
t_fall	Output Fall Time	N/A	76.1 ps

7.3 Maximum Operating Frequency (Fmax)

The ultimate operating speed limit of the LFSR is bound by its critical physical path,

$$T_{min} = tcq + tXOR + tsu$$

$$T_{min} = 369.2 \text{ ps} + 124.3 \text{ ps} + 78.5 \text{ ps} = 572.0 \text{ ps}$$

$$F_{max} = 1 / T_{min} = 1 / 572.0 \text{ ps} \approx 1.75 \text{ GHz}$$

7.4 Power and DC Signal Integrity

Power profiling was executed by integrating the transient current from the primary supply during active high-frequency PRBS switching states.

Parameter	Description	Value
Dynamic Power	Average power consumed	246.1 μW
Static Power	Leakage power (Clock = 0V)	18.4 nW
V_OH	Measured Output High	1.812 V
V_OL	Measured Output Low	0.012 V
Physical Area	Total Silicon Area	7855.28 μm²

8. Conclusion

This project successfully demonstrates the full-custom, transistor-level physical implementation of a 10-bit Linear Feedback Shift Register (LFSR) utilizing the SCL 180nm CMOS process node. By bypassing automated synthesis in favor of a strict static-CMOS 6-NAND topology, the design achieved highly predictable drive strengths and eliminated the clock-skew vulnerabilities inherent to standard pass-transistor logic.

The critical LFSR "Zero-Lock" vulnerability was permanently resolved at the hardware level. By custom-designing asynchronous **Dff7474_prst** and **Dff7474_rst** standard cells, a deterministic **1100000000** initialization seed was successfully hardwired into the silicon layout, completely avoiding the parasitic delay penalties associated with datapath multiplexers.

Strict backend-of-line (BEOL) routing disciplines—including Euler path active-area sharing, pitch-aligned cell abutment, and orthogonal metal isolation—were executed to minimize physical RC degradation. The final extracted IP macro occupies a minimal silicon footprint of **7855.28 μm^2** . Post-layout PEX simulations confirmed the physical robustness of the datapath, achieving a maximum operating frequency (Fmax) of **1.75 GHz** while maintaining a low dynamic power consumption of **246.1 μW** at a **1.8V nominal supply**. The extracted macro is 100% DRC* and LVS clean, explicitly optimized for dense, high-frequency integration in larger SoC architectures. (DRC* = Ignored Density Errors)

9. References

[1] Fairchild Semiconductor, "DM74LS74A Dual Positive-Edge-Triggered D Flip-Flops with Preset, Clear and Complementary Outputs," Product Datasheet, Revised March 2000. (Reference for the 6-NAND dual-control sequential baseline architecture).

[2] J. Koeter, "What's an LFSR?," Texas Instruments, Application Report SCTA036A, Dec. 1996. (Reference for maximal-length pseudo-random pattern generation, tapping polynomials, and deterministic seeding logic).